MIL-F-46488B
16 MARCH 1981
SUPERSEDING
MIL-F-46488A
15 June 1963

MILITARY SPECIFICATION

FUZE, BASE DETONATING, M534A1 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Loading, Assembling and Packing for one type of artillery fuze designated as Fuze, Base Detonating, M534A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Test For
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-453 - Radiographic Inspection
MIL-STD-650 - Explosive Sampling, Inspection and Testing

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8860724 - Fuze, Base Detonating, M534A1 Loading Assembly
8849028 - Carton, Packing, Ammunition for Fuze, BD, M534A1
8849029 - Box, Packing, Ammunition for Fuze, BD, M534A1

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Fuze assembly.-The fuze assembly shall comply with all requirements specified on Drawing (dwg.) 8860724 and with all requirements specified in the applicable specifications.

3.3 Specific gravity of booster pellet.-The specific gravity of the booster pellet shall be 1.60 plus 0.05 grams per cubic-centimeter (gm/cc) when tested as specified in 4.5.1.

3.4 Jolt and jumble.-The fuze assembly shall be safe to handle in accordance with the requirements of MIL-STD-331 when tested as specified in 4.5.2.

3.5 Waterproofness.-When immersed in water as specified in 4.5.3, disassembled fuzes shall show no evidence of moisture on any interior surface.

3.6 Radiographic examination.-The fuze assembly when radiographed and examined as specified in 4.5.5 shall exhibit no evidence of the following.

3.6.1 Fuze assembly.-The fuze assembly shall be examined for: the shutter not being in the unarmed position; missing or mal-assembled components; and any other condition differing from the normal assembled condition as shown on dwg. 8860724.

3.6.2 Booster pellet.-The booster pellet shall not show a radiographic projection of a low density area or disjuncture lying outside the bands (zones) specified in 4.5.5.2.1.

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3.6.3 Radiographic quality.-Radiographs taken for booster pellet, armed fuze and missing component evaluation shall be taken in accordance with MIL-STD-453 and shall be of a quality which will permit proper evaluation of the fuze assembly acceptability. Minimum optical density of each radiograph in the area being inspected shall not be less than 1.25 and shall be a density range that will permit identification of defects.

3.7 Proving ground tests.-These tests shall be performed as specified in 4.5.4.

3.7.1 Arming safety.-The fuze, when assembled into projectiles, shall not function upon impact with a four (4) inch (10.2 centimeters (cm)) plywood target placed sixteen feet (4.88 meters (m)) from the muzzle of the weapon.

3.7.2 Functioning.-The fuze, when assembled into projectiles, shall function upon impact with a four inch (10.2 cm) plywood target placed one hundred fifty feet (45.7 meters) from the muzzle of the weapon.

3.8 Data requirements.-The contractor shall generate data in accordance with the requirements of the data item description cited in 4.4.2.b.

3.9 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following.

3.10.1 General.-The fuze assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter. The cleaning method used shall not be injurious to any component; nor, shall parts be contaminated by the cleaning agent.

3.10.2 Booster loading.-The interior surface of the booster cup shall be clean and dry when the lead charge and booster charge (pellet) are loaded into it. The threads shall be free of explosive and foreign matter. If a chemical cleaner is required, no solvent other than acetone shall be used; nor, shall the solvent be allowed to come into contact with either charge.

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3.10.2.1 Surface breakout and chipping.-The booster pellet shall be free of breakout greater than three-thirty seconds inch (3/32 inch). When the pellet is preformed prior to assembly to the booster cup, the edges of the pellet shall not be chipped greater than one-sixteenth inch (1/16 inch) deep.

3.10.3 Packing and marking.-Packing materials shall be thoroughly dry when used and shall be assembled in accordance with the applicable drawings and specifications. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

Part Description	Drawing	Quantity
Disc, Closing	8860732	25
Pellet, Booster	8860730	35
Booster Cup, Prior to Loading Booster Pellet	8860724	25
Booster Cup, Prior to Assembly To Fuze Body	8860724	25
Fuze, Base Detonating M534A1 Loading Assembly	8860724	261

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Fuze, Base Detonating, M534A1				See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Closing Disc (Dwg. 8860732)				
	Examination for Defects	25		3.1	4.4.2.1
	Booster Pellet (Dwg. 8860730)				
	Examination for Defects	25		3.2	4.4.2.2
	Specific Gravity	10		3.3	4.5.1
	Booster Cup, Prior to Loading Booster Pellet (Dwg. 8860724)				
	Examination for Defects	25		3.2	4.4.2.3
	Booster Cup, Prior to Assembly to Fuze Body (Dwg. 8860724)				
	Examination for Defects	25		3.2	4.4.2.4
	Fuze, Base Detonating, M534A1 Loading Assembly (Dwg. 8860724)				
	Examination for Defects	25		3.2	4.4.2.5
	Jolt and Jumble Test	36		3.4	4.5.2
	Water Proofness	50		3.5	4.5.3
	Radiographic Inspection	261		3.6	4.5.5
	Ballistic Test	150		3.7	4.5.4

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Fuze metal parts of one lot interfix number from one manufacturer.
b. Composition A-5 of not more than two consecutively numbered lots from one lot interfix number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Inspection Level II.
b. Defect reporting.-The contractor shall prepare a monthly report of all critical defects found, including the number of components/assemblies inspected and corrective action taken, in accordance with Data Item Description DI-R-1721 and decline the reports as specified in the contract.
c. Visual inspections.-When compliance with the applicable requirements is in doubt as a result of visual inspection, these characteristics may be measured or gaged to determine acceptability. In the event of a dispute, four power (4X) magnification shall be used in performing visual inspections for such defects as burrs and foreign matter.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Disc, Closing				8860732
					NEXT HIGHER ASSEMBLY
					8860724
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Diameter		0.65%	3.1	Gage
202	Thickness		0.65%	3.1	Gage
203	Evidence of poor workmanship		0.65%	3.10.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Pellet, Booster		8860730
			NEXT HIGHER ASSEMBLY
			8860724

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pellet cracked				
Major					
101	Specific gravity	a.		3.3	Test/4.5.1
Minor					
201	Length		0.65%	3.2	Gage
202	Diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.10.2	Visual

NOTES: a. See 4.4.3.1 for sampling plan.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Booster Cup, Prior to Loading Booster Pellet				8860724
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Lead charge above flush			3.2	Gage
Major					
101	Lead charge below flush, max.		100%	3.2	Gage
102	Lead charge missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Booster Cup, Prior to Assembly to Fuze Body		8860724
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Flatness of rear face of booster cup		100%	3.2	Gage
102	Booster pellet above flush		0.40%	3.2	Gage
103	Explosive or foreign matter on thread or rear face		0.40%	3.2	Visual
104	Booster pellet missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Fuze, Base Detonating, M534A1 Loading Assembly		8860724
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Radiographic inspection			3.6	4.5.5/Test
2	Radiographic inspection acceptance mark missing			3.2	Visual
3	Waxed felt disc missing			3.2	Visual
4	Jolt and jumble test	a		3.4	4.5.2/Test
Major					
101	Adhesive missing from any thread		0.40%	3.2	Visual
102	Booster cup fails to withstand reapplication of minimum assembly torque with adhesive wet		0.40%	3.2	Gage
103	Locking ring fails to withstand reapplication of minimum assembly torque with adhesive wet		0.40%	3.2	Gage
104	Booster cup fails to withstand reapplication of minimum assembly torque with adhesive wet		0.40%	3.2	Gage
105	Alignment screw above flush		0.40%	3.2	Gage
106	Length, max.		0.40%	3.2	Gage
107	Waterproofness test	b		3.5	4.5.3/Test
Minor					
201	Alignment screw staking missing or improper		0.65%	3.2	Visual
202	Marking missing incorrect or illegible		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:
a. See 4.4.3.2 for sampling plan.
b. See 4.4.3.3 for sampling plan.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Carton, Packing Prior to Sealing		8849028
			NEXT HIGHER ASSEMBLY
			8849029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Separator damaged to the extent fuzes touch each other		0.40%	3.2	Visual
102	Any packing component missing		0.40%	3.2	Visual
Minor					
201	Fuze missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Sealed Packing Carton		8849028
			NEXT HIGHER ASSEMBLY
			8849029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%	3.2	Manual
102	Marking missing, incorrect or illegible		0.40%	3.2	Visual
Minor					
201	Carton torn, cut or punctured exposing contents		0.65%	3.2	Visual
202	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal box completely		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-46488B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Sealed Packing Box				8849029
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Box broken, loose board or protruding nail		0.40%	3.2	Visual/Manual
102	Strapping missing broken or loose		0.40%	3.2	Visual/Manual
103	Marking missing, incorrect or illegible		0.40%	3.2	Visual
Minor					
201	Strapping improperly engaged or mislocated		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
203	Evidence of poor workmanship		0.65%	3.10.2	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Specific gravity of Comp A-5 booster pellet, Major Defect.-Two booster pellets shall be randomly selected from each eight hours production for this test. If any sample fails to comply with the requirement of dwg. 8860724, the production represented by the sample shall be rejected. This test shall be conducted as specified in 4.5.1.

4.4.3.2 Jolt and jumble, Critical Defects.-These tests shall be performed on the fuze assemblies selected for the tests as specified in 4.5.2. If one or more fuze assemblies fail to meet the acceptance criteria herein, the lot shall be rejected.

4.4.3.2.1 Initial production.-Beginning with the first fuze lot produced and continuing until three consecutive lots have complied with the acceptance criteria herein, a sample of thirty-six fuze assemblies shall be selected from each lot for these tests.

4.4.3.2.2 Subsequent production.-After three consecutive lots have met the criteria specified in 4.4.3.2.1, a sample of twelve fuze assemblies shall be selected from each lot for these tests.

4.4.3.3 Waterproofness.-This test shall be performed on the fuze assemblies selected for the tests as specified in 4.5.3.

4.4.3.3.1 Initial production.-Beginning with the first fuze lot produced and continuing until three consecutive lots have complied with the acceptance criteria herein, fifty fuze assembly samples shall be selected from each lot for this test. If two or more fuze assemblies show evidence of moisture upon disassembly, the lot shall be rejected.

4.4.3.3.2 Subsequent production.-After three consecutive lots have complied with the criteria of 4.4.3.3.1, fifteen sample fuze assemblies shall be selected from each lot for this test. If one or more fuze assemblies show evidence of moisture upon disassembly, the lot shall be rejected.

4.4.3.4 Proving ground tests

4.4.3.4.1 Arming safety, Critical Defect.-This test shall be performed as specified in 4.5.4.1.

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4.4.3.4.1.1 Initial production.-Beginning with the first fuze assembly lot produced and continuing until three consecutive lots have complied with the acceptance criteria herein, seventy-five fuze assemblies shall be tested for arming safety. If one or more fuze assemblies function upon impact with the target, the lot shall be rejected.

4.4.3.4.1.2 Subsequent production.-After three consecutive lots have complied with the criteria of 4.4.3.4.1.1, ten fuze assemblies shall be tested for arming safety. If one or more fuze assemblies function upon impact with the target, the lot shall be rejected.

4.4.3.4.2 Functioning, Major Defect.-This test shall be performed as specified in 4.5.4.2.

4.4.3.4.2.1 Initial production.-Beginning with the first fuze assembly lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, seventy-five fuze assemblies shall be tested for functioning. If two or more fuze assemblies fail to function upon impact with the target, the lot shall be rejected.

4.4.3.4.2.2 Subsequent production.-After three consecutive lots have complied with the criteria of 4.4.3.2.1, ten fuze assemblies shall be tested from each lot for functioning. If two or more fuze assemblies fail to function upon impact with the target, the lot shall be rejected. If one fuze assembly fails to function, then an additional sample of sixty-five fuze assemblies shall be tested for functioning. If no failures are obtained from the additional sample, then the lot shall be accepted.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. (See 6.2 and Section 6 of MIL-A-48078).

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5 Test procedures

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4.5.1 Specific gravity of Comp A-5 booster pellet.-The specific gravity of the booster pellet shall be determined in accordance with Method 203.1 of MIL-STD-650.

4.5.2 Jolt and jumble.-The fuze assemblies selected for these tests shall first be subjected to the Jolt Test in accordance with Method 101.1 of MIL-STD-331, followed by the Jumble Test in accordance with Method 102.1 of MIL-STD-331. Each fuze assembly shall then be disassembled and examined for the defects listed below.

a. Locking segment and striker needle assembly.-The locking segment and striker needle assemblies shall be examined for evidence of one or more open or inoperable segments, excessive striker needle length, and inoperable creep springs.
b. Arming delay assembly.-The arming delay assemblies shall be examined for evidence of the shutter not being in the locked, unarmed position. This position is defined as the condition in which the safety plunger remains in physical contact with the locking plate which prevents any movement of the shutter assembly.

Fuze assemblies subjected to these tests shall not be returned to the lot and shall be disposed of in accordance with local approved procedures.

4.5.3 Waterproofness.-The fuze assembly samples selected for the waterproofness test shall be tested in accordance with the procedure specified in Method 108 of MIL-STD-331, except that the vessel containing the samples shall not be pressurized. Upon their disassembly, each fuze assembly shall be inspected as specified in 4.5.2, preceded by the check for the presence of moisture as prescribed by MIL-STD-331. If the fuze components have dried in the time interval from their disassembly and examination, a water atomizer is to be used to enhance the reflection of any salt stains which may be present when subjected to an ultraviolet source.

4.5.4 Proving ground tests.-The fuze samples which are to be ballistically tested as prescribed herein shall be fired in a M68 gun. The ammunition in each phase shall be conditioned to $70^{\circ} \pm 2.5^{\circ}F$ ($21.1^{\circ} \pm 1.5^{\circ}C$) until all propellant is stabilized at a uniform temperature prior to firing, which normally requires a period of twenty-four hours, minimum (see 6.3).

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4.5.4.1 Arming safety phase.-The projectiles in which the fuzes have been assembled shall be fired at a four inch (ten and two tenth centimeters) thick plywood target located sixteen feet (four and nine-tenths meters) from the muzzle of the gun. Observation shall be made for the non-functioning of the fuze prior to and on impact with the plywood target. Functioning, if any, of the fuze subsequent to impact with the plywood target shall be recorded for informational purposes on the firing record.

4.5.4.2 Functioning phase.-The projectiles containing the fuze samples shall be fired at a four inch (ten and two-tenths centimeters) thick plywood target located at a distance of one hundred fifty feet (forty-five and seven tenths meters) from the muzzle of the gun. Observation shall be made for fuze functioning upon of immediately after projectile impact with the plywood target. Functioning subsequent to impact with the plywood target shall be noted on the firing record, for information, for the fuzes which fail to function (dud) as specified herein.

4.5.4.3 Loading and assembly of fuze assemblies.-The fuze assembly samples shall be assembled to WP-filled or inertly loaded M416 projectiles. HE loaded M393A1 projectiles may be used only with specific instructions from the Contracting Officer.

4.5.5 Radiographic examinations.-Radiographic examination of fuzes shall be performed in accordance with the general requirements of MIL-STD-453; as specified herein; and, with specific operating procedure prepared by the contractor, which shall be submitted to and approved by the Contracting Officer prior to the submission of the first article sample. No flouroscopic or other non-film method of radiographic imaging shall be substituted for film radiography prior to specific review and approval by the Contracting Officer. In general, any such method must meet all radiographic image quality and sensitivity requirements of this specification, as well as, other required characteristics inherent to film imaging. An identification numbering system shall be employed which shall provide direct, unambiguous correlation between each fuze and its radiographic image; and, which shall be used for reporting purposes.

4.5.5.1 Radiographic exposure of fuzes.-Prior to exposure to radiation, each fuze shall be dropped, base down, from a height of one inch, minimum, in order to dislodge loose components. Fuzes shall be positioned in a manner such that complete coverage of one or more fuzes is provided on one film. The alignment of each fuze shall be controlled such that the radiation source shall be in a plane perpendicular to the longitudinal axis of each fuze, intersecting the fuze at its center. Orientation of each fuze shall register on the external set screw, such that the set screw shall project onto the central axis of the fuze radiographic image.

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Source to film distance shall not be less than four feet and peak radiation energy shall not be less than three hundred kilovolts. The radiographic technique shall be maintained to consistently produce the image quality characteristics displayed on the approved standard reference radiographs obtained in accordance with the radiographic sensitivity procedures specified herein.

4.5.5.1.1 Sensitivity.-The sensitivity of the radiographic technique for fuze inspection shall be determined by an examination of internal fuze features. Two or more standard reference radiographs of fuzes in the required orientation shall be produced and used for comparison. Reference radiographs shall be submitted to the Contracting Officer for review and approval prior to the submission of the first article sample, and one of the approved reference radiographc will be retained for the Contracting Officer by the technical activity. The radiographic sensitivity shall not be considered satisfactory unless all major internal features on every fuze image are as sharply defined as on the approved reference radiographs. Failure to meet this sensitivity requirement on any fuze shall require the re-radiographing of that fuze and a diagnosis of that particular failure.

4.5.5.1.2 Review of radiographic sensitivity and techniques.-The required reference radiographs of fuzes, as well as, radiographic diagrams and techniques shall be submitted for review and approval to the Contracting Officer, prior to the submission of the first article sample. One of the initial radiographs will be retained at the technical activity for the Contracting Officer. One radiograph of fuzes shall be submitted for review and comparison to the retained reference radiograph once each month for the first three months of production on each contract; and, once every three months thereafter for the duration of each contract. All radiographs submitted shall be produced using materials and techniques identical to those used on normal production lines.

4.5.5.2 Examination of fuze radiographs.-Each fuze radiograph shall be viewed for the following defects listed as Critical in nature: Loaded shutter not in the unarmed position (see dwg. 8860724); Delay assembly missing or malassembled; Creep spring missing; and, Closure disc missing. The following defects are listed as Major in nature: One or more segments missing or improperly assembled; Booster pellet cracked or containing voids (see 4.5.5.2.1); and, Any component missing, damaged, or malassembled. Following their examination, each acceptable fuze shall be marked with indellible ink on its base as required by dwg. 8860724, denoting passage of the radiographic examination. Any fuze determined to be defective shall be re-radiogaphed for the verification of the defects or defects. Fuzes with verified critical defects may be disposed of locally following immediate notification of the discovery to the Contracting Officer.

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4.5.5.2.1 Interpretation of explosive defects.-Any radiographic projection of a low density area or disjuncture in the booster pellet which extends through the boundary of any band in the projected image of the booster pellet as described herein shall be cause for the rejection of a fuze. The preparation and use of a suitable template for use in this determination is suggested.

4.5.5.2.1.1 Disjuncture.-When the booster pellet is prepressed in two equal parts as allowed by dwg. 8860730, a radiographic projection of a disjuncture in the booster explosive charge will only be allowed to appear at the juncture of the two pellets. The maximum allowable width of a radiographic projection of a disjuncture between pellets shall be one-sixteenth inch (0.0625 inch), and shall only lie within a three-sixteenths inch (0.1875 inch) wide band centered at the junction of the pellets.

4.5.5.2.1.2 Low density areas.-The radiographic projection of a low density area will only be allowed to appear in three bands (zones) within the booster charge. Two bands shall be of one-eighth inch (0.125 inch) maximum width at the top and bottom (one band each end) of the booster charge. The remaining band shall be located in the booster charge and specified for disjunctures in 4.5.5.2.1.1 above. The total aggregation of low density areas shall not exceed one-sixteenth square inch in any band.

4.5.5.3 Reliability of examination.-As a means to maintain the reliability of the examination of fuze radiographs, fuzes containing known defects shall be randomly introduced into inspection lots of fuzes at a rate not less than once every five hundred fuzes. These known defective fuzes shall only differ from production fuzes in that they shall contain only one purposeful defect and shall be marked in such a manner as to assure that they cannot be mistaken for production fuzes and subsequently released to field or industrial stocks. Furthermore, these fuzes shall have their detonators replaced or modified in a manner which shall not be discernible on the subsequent radiographs of these known-defective fuzes. No more than one known-defective fuze shall be radiographed with unknown fuzes at any one time; and, the frequency and time of the introduction of these known-defective fuzes into the radiographic inspection process shall be varied to preclude radiographic inspectors (viewers) from examining only for the known-defective fuzes. If a known-defective fuze is not identified by the viewer(s), one hundred percent reverse order screening of the fuze radiographs viewed since the last known defective fuze shall be made.

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4.5.5.4 Viewer certification.-Inspection personnel (viewers) required for this radiographic examination shall be certified as qualified as Level II personnel in accordance with the requirements of MIL-STD-410.

4.5.5.5 Radiographic film.-Only an Industrial Radiographic Film of an approved type shall be used to radiograph this fuze. Substitutes may be used only after evaluation and approval by the technical activity. The following are initially approved types:

a. Eastman Kodak Company, Types AA, T and M.
b. E.I. Dupont De Mours & Co., Types NDT-55, NDT-65 and NDT-70.
c. GAF-800.
d. GAF-400.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-The fuze assemblies shall be preserved and packaged Level B in accordance with dwg. 8849028.

5.2 Packing.-The packaged fuze assemblies shall be packed Level B in accordance with dwg. 8849029.

5.3 Marking.-Marking shall be as specified on dwg. 8849029.

6. NOTES

6.1 Ordering data

6.1.1 Procurement requirements.-See MIL-A-48078.

6.1.2 Data requirements.-Deliverable data required by the specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.4.2.b	Quality Inspection Defect Report	DI-R-1721

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

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6.1.2.1 Delivery of data to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for delivery of one copy, each, to the technical activity, of the Ammunition Data Cards and Quality Inspection Defect Reports prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

6.2 Submission of designs for approval.-Submit equipment designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.3 Firing record information.-The ballistic acceptance test firing records shall include the serial numbers of all weapon and fire control components. Stock component lot numbers of warmers and spotters shall also be included. For each round fired, the record shall contain deflection, elevation, time of firing, gun tube round number, manner of functioning (if applicable) pertinent flight characteristics and meteorological data. The following addresses are to be added to the Ballistic Test Request (BTR) for the distribution of firing records to the technical activity: Commander, US ARmy Armament Research and Development Command, ATTN: DRDAR-QAR-E and DRDAR-LCU-CV, Dover, New Jersey 07801.

6.4 Proving ground test summary

Test Phase	Initial Production	Subsequent Production	Sampling Paragraph	Procedure Paragraph
Arming Safety	75	10	4.4.3.4.1	4.5.4.1
Functioning	75	10*	4.4.3.4.2	4.5.4.2

*If one fuze sample fails to function, an additional sample of sixty-five (65) fuzes are tested for functioning.

6.5 Changes from previous issue.-Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR

Project Number: 1390-A227

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